

Dr. Markus Bauer

State: 2026-05-15

[ORCID](#) | [Google Scholar](#)

[GitHub](#) | [Stackoverflow](#)

[LinkedIn](#)

Homepage: <https://markus1bauer.github.io/>

Professional Experience

- Since 08/2023 [Postdoc \(75%\)](#): Chair of Restoration Ecology (TUM)
- Since 06/2022 [Self-employed vegetation mapper \(25%\)](#): Landschaftsökologie Bauer, Freising
Biotope mapping, Bavarian Compensation Ordinance, vegetation monitoring
- 01/2019–07/2023 [Research assistant](#): Chair of Restoration Ecology (TUM)
Development of a monitoring concept, a restoration substrate with bricks

Education

- 06/2021–03/2024 [Dr. rer. nat.](#) Restoration Ecology
Technical University Munich

Dissertation: [Seed–substrate combinations for restoration of urban and dike grasslands under environmental fluctuations](#)
Supervisor: Prof. Dr. Johannes Kollmann
- 10/2016–12/2018 [M.Sc.](#) Environmental planning and ecological engineering (Project studies)
Technical University Munich

Thesis: [Monitoring of the Garchinger Heide. Vegetation ecological studies of a reference ecosystem under changed environmental conditions over 34 years](#)
Chair for Restoration Ecology
- 10/2012–10/2016 [B.Sc.](#) Landscape architecture and planning (Project studies)
Technical University Munich

Thesis: [Zweites Monitoring von Rohbodenstandorten des Inns nach Renaturierungsmaßnahmen](#); Chair for Restoration Ecology

Stays abroad

- 2025
(3.5 months) Sabbatical: Restoration Ecology Lab, Michigan State University, USA
Visited Professor: Lars Brudvig | Experiment: Tall grass prairie restoration
- 10/2014–02/2015
(4 months) Internship abroad: Botanical Garden, León, Nicaragua
Construction of butterfly house, plant care, guided tours
- 07/2014
(1 month) Vegetation ecology excursion: Kyrgystan and Kazakhstan
Guide: Professor Jörg Pfadenhauer (Chair of Restoration Ecology)
- 03–04/2014
(2 months) Internship abroad: Rooiklip Research Camp (TUM), Namibia
Seed predation experiments with *Crotalaria podocarpa*
- 2013–2026
(each 3–10 days) Multi-day vegetation ecology excursions: Lake Neusiedel, Sardinia,
Tagliamento, Alps, Luxemburg, Colorado, Northern Spain

Research Projects

- 2025–2027 Dike experiment at River Inn: Testing substrate–seed combinations for targeted greening (VERBUND Innkraftwerke)
- 2023–2025 Grassworks: What works in grassland restoration in Germany, and why? (FEdA of the Federal Ministry of Education and Research)
- 2021–2025 River dikes at the Danube: Integrated flood protection and biodiversity conservation – the potential of innovative dike greening (DBU, WIGES GmbH)
- 2021–2022 Vegetation monitoring of the nature conservation area Garchinger Heide: Monitoring concept and first survey (Heideflächenverein Münchner Norden)
- 2019–2020 Brick substrates: Development of technologies for reusing waste bricks as a functionalized growing medium with storage properties (e.g. moisture, nutrients), for example for landfill capping and as a substrate for tree planting. (ZIM project of the Federal Ministry for Economic Affairs, ZF4025032SA8)

Selected Publications (Mentored authors: students, PhDs)

- 2026 Appeltauer S, Knöppler M, Teschauer M, Kollmann J & **Bauer M** ([2026](#)) The restoration debt remains: Topsoil removal and different mowing regimes reach desired states of calcareous grassland but do not match the reference. – *Applied Vegetation Science* 29, e70067.
- 2024 **Bauer M**, Huber J, Kollmann J ([2024](#)) Beta diversity of restored dike grasslands is strongly influenced by uncontrolled spatio-temporal variability. – *Journal of Vegetation Science* 35, e13293.
- 2023 **Bauer M**, Huber J, Kollmann J ([2023](#)) Fit by design: Developing substrate-specific seed mixtures for functional dike grasslands. – *Journal of Applied Ecology* 60, 2413–2424.

- 2023 Teixeira LH*, **Bauer M***, Moosner M, Kollmann J (2023) River dike grasslands can reconcile biodiversity and different ecosystem services to provide multifunctionality. – *Basic and Applied Ecology* 66, 22–30
- 2020 **Bauer M**, Albrecht H (2020) Long-term vegetation change in a 100-year-old calcareous grassland reserve in Germany. – *Basic and Applied Ecology* 42, 15–26.
- Since 2016 All papers (see [Google Scholar](#))

German Publications

- 2026 **Bauer M** (2026) Prärierenaturierung und Waldbrände – als Posdoc in den USA. – *Nodium des Alumni-Clubs Landschaft TUM* 18, 32–33.
- 2023 **Bauer M**, Huber J, Kollmann J (2023) Fluktuierende Vegetationsbestände. Eine Herausforderung für die naturschutzfachliche Bewertung. – *Nodium des Alumni-Clubs Landschaft TUM* 15, 106–107.
- 2020 **Bauer M**, Kiefer J, Albrecht H (2020) Vegetationsveränderungen auf Dauerbeobachtungsflächen im Naturschutzgebiet Garchinger Heide. – *Berichte der Bayersichen Botanischen Gesellschaft* 90, 43–66.

Research Conferences

- | | | | |
|------------|--|--------------------|--------|
| 2026 | Conference on assisted migration (Federal Agency of Island Vilm, GERTalk Nature Conservation, BfN) | | |
| | IAVS (International Association of Vegetation Science) | Gijón, ESP | Talk |
| 2025 | ESA (Ecological Society of America) | Baltimore, MY, USA | Talk |
| | IAVS (International Association of Vegetation Science) | Greeley, CO, USA | Talk |
| 2024 | GfÖ (Ecological Society of GER, AUT and SUI) | Freising, GER | Talk |
| | SERe (Society of Ecological Restoration Europe) | Tartu, EST | Talk |
| 2023 | GfÖ (Ecological Society of GER, AUT and SUI) | Leipzig, GER | Poster |
| 2022 | GfÖ (Ecological Society of GER, AUT and SUI) | Metz, FRA | Talk |
| | IAVS (International Association of Vegetation Science) | Madrid, ESP | Talk |
| 2021 | PopBio (Plant Population Biology group of the GfÖ) | Online | Poster |
| | SER (Society of Ecological Restoration World) | Online | Talk |
| 2019 | GfÖ (Ecological Society of GER, AUT and SUI) | Münster, GER | Talk |
| | IAVS (International Association of Vegetation Science) | Bremen, GER | Poster |
| 2017 | SERe (Society of Ecological Restoration Europe) | Freising, GER | – |
| Since 2019 | FlorSoz (Luxemburg, Rostock, Freising (guide), Koblenz, Oldenburg) | | |

Grants

- 2025–2027 Research (F+E) project for grassland restoration on dikes (VERBUND Innkraftwerke GmbH; 219k EUR)

2021	Vegetation monitoring of the nature conservation area Garchinger Heide: Monitoring concept and first survey (Heideflächenverein Münchner Norden; 24k EUR)
2021–2024	Doctoral Scholarship (Deutsche Bundesstiftung Umwelt, DBU; 62k EUR)

Awards

2024	Award for the best dissertation at the TUM. Awarded by the TUM president Prof. Thomas Hofmann for the Johannes B. Ortner foundation
2020	Best Master's thesis of the department landscape architecture and planning. Awarded by the mayor of the city Freising Tobias Eschenbacher

Invited Presentations

2027-02-10	Bavarian Botanical Society, Munich: Winter lecture series
2026-04-28	Federal Agency on Nature Conservation (BfN), Island Vilm: Workshop on Assisted Migration
2025-12-15	University of Bozen-Bolzano: Conference on Restoration under Climate Change
2025-09-05	Michigan State University: Plant Biology Seminar
2018-12-18	Bavarian Botanical Society, Munich: Winter lecture series

Teaching

Lectures and Seminars	Lecture Population and Community Ecology Seminar Conservation and Community Ecology
Courses	Block course Analysis of Ecological Data
Excursions	Bavarian Forest Mountain forest at Benediktenwand (Alps) Floodplain forests at estuary of river Isar Beech forests of Kelheim and Siegenburg Garchinger Heide
Teacher training	Training course for postdocs: Foundations of teaching – Basic-level certificate of learning and teaching in higher education of the Bavarian Universities, ProLehre, Department for Teaching and Learning in Higher Education, Technical University of Munich (TUM)

Supervision/Mentoring

PhDs	Max Thiele (since 2025, co-supervisor)
Master's thesis	Since 2025: 1 completed, 4 ongoing
Bachelor's thesis	Since 2022: 1 completed

Student projects and internships Master's landscape planning projects (4 groups)
Experimental Ecology (3 groups)
Research internships (2 students)

Service

Journal Referee Restoration Ecology (6), Flora (2), Journal of Vegetation Science (2), Applied Vegetation Science (1), Perspectives in Plant Ecology, Evolution and Systematics (1)

Engagement current **Advisory Board of Freising for Nature Conservation** (since 2024, member)
[Advisory Board of „Heideflächenverein Münchner Norden e.V.“](#) (since 2024, member for the Technical University Munich. This is an association for the management of several conservation areas between Munich and Freising)
Wikipedia (since 2022, active author for ecological topics)

Engagement past Alumni-Club Landschaft TUM ([ACL](#), Freising; secretary 2020–2025; editor of the club journal 2019–2023)

Memberships International Association of Vegetation Science ([IAVS](#), since 2023)
Ecological Society of Germany, Austria and Switzerland ([GfÖ](#), since 2023)
Floristisch-Soziologische Arbeitsgemeinschaft ([FlorSoz](#), since 2019)
Bavarian Botanical Society ([BBG](#), since 2019)
Alumni-Club Landschaft TUM ([ACL](#), since 2019)

Skills

Languages English (C1), Spanish (A2), French (A2)

Computer [Frequentist](#) and [Bayesian](#) statistics and [data wrangling](#) with R (very good)
Package development with R
[Shiny with R](#) (advanced)
[GIS with R](#) and QGIS (good)
GitHub (good)
MS Office (good), MS Access (advanced)

License Driving license B, Betrieblicher Ersthelfer (seit 2024)